We stress that the implementation of this method is in fact quite straightforward, despite its apparent complexity. The following details regarding the implementation are discussed in [220]:

- For the integration of the Nosé terms, it is recommended to use a further splitting:

$$\mathcal{L}_{\mathrm{N}} = \mathcal{L}_{\mathrm{N}}^{(1)} + \mathcal{L}_{\mathrm{N}}^{(2)},$$

where

$$\mathcal{L}_{\mathrm{N}}^{(1)} = \frac{1}{2}\frac{\mu_1\xi_1^2\xi_2 p}{\Lambda}\frac{\partial}{\partial p} + \frac{1}{2}\frac{\mu_1\xi_1^3\xi_2}{\Lambda}\frac{\partial}{\partial \xi_1} - \xi_2\xi_1\frac{\partial}{\partial \xi_1},$$

and

$$\mathcal{L}_{\mathrm{N}}^{(2)} = \mu_2^{-1}[\mu_1\xi_1^2 - \beta^{-1}]\frac{\partial}{\partial \xi_2}.$$

Moreover a Yoshida/Suzuki type higher order decomposition is used to obtain more accuracy in this component.

- The term $\exp[h\mathcal{L}_F]$ (like similar terms) is completely integrable. Note that in general this reduces to solving

$$\dot{p} = F - p^2 F/\Lambda,$$
$$\dot{\xi}_1 = p\xi_1 F/\Lambda,$$

with F a constant.

The solution in a timestep h is

$$p(h) = \frac{p(0) + (F/m)s(h)}{\dot{s}(h)},$$
$$\xi_1(h) = \frac{\xi_1(0)}{\dot{s}(h)},$$

where

$$s(t) = b^{-1/2}\sinh(t\sqrt{b}) + \frac{a}{b}(\cosh(t\sqrt{b}) - 1)$$

with $a = Fp(0)/\Lambda$ and $b = F^2/\Lambda$, $\Lambda = \beta^{-1}$.

Care must be taken in the evaluation of s and $\dot{s}$ for a, b close to zero.

Note that other decompositions are possible [220].